



Redes sociales y salud mental

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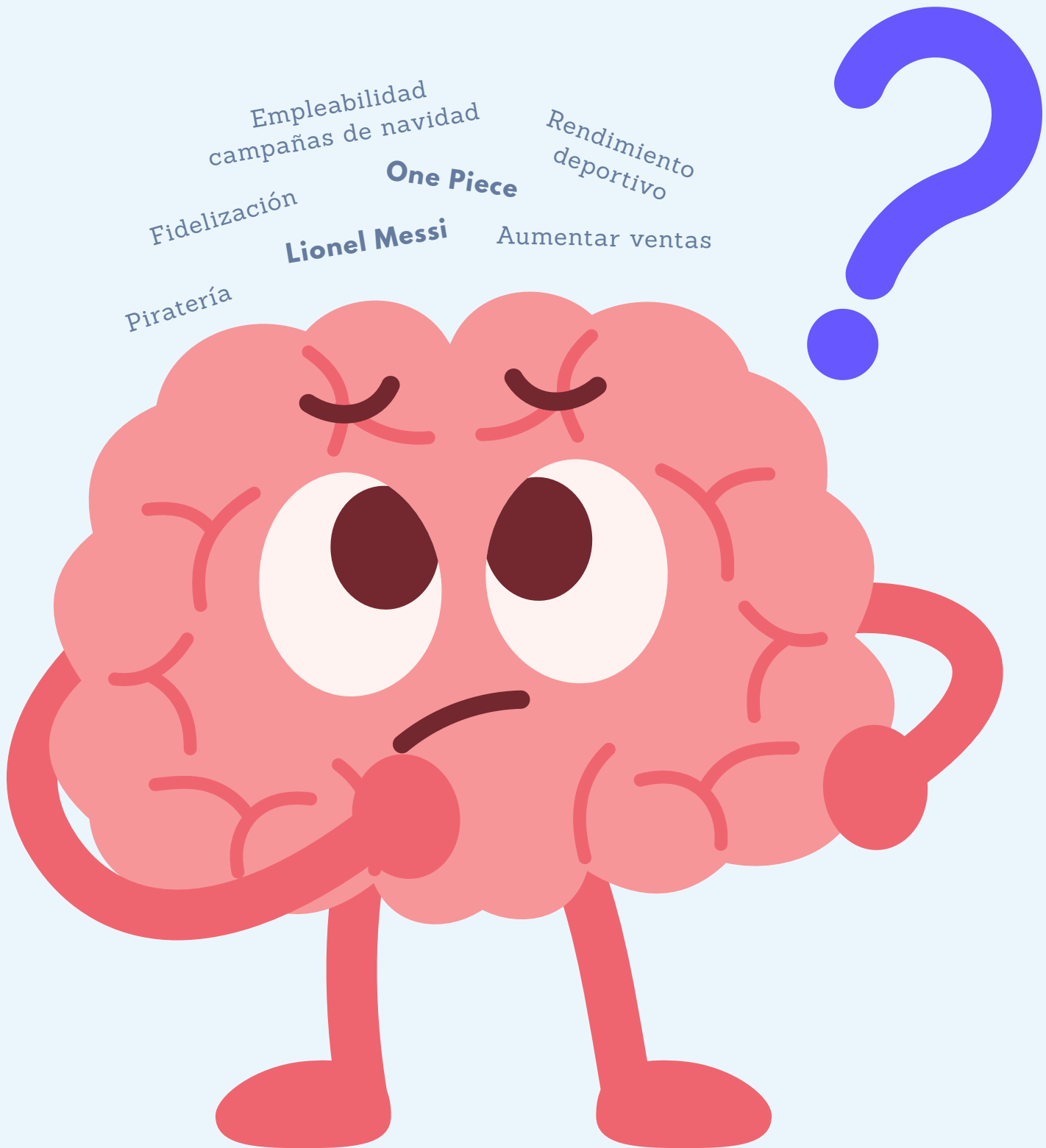


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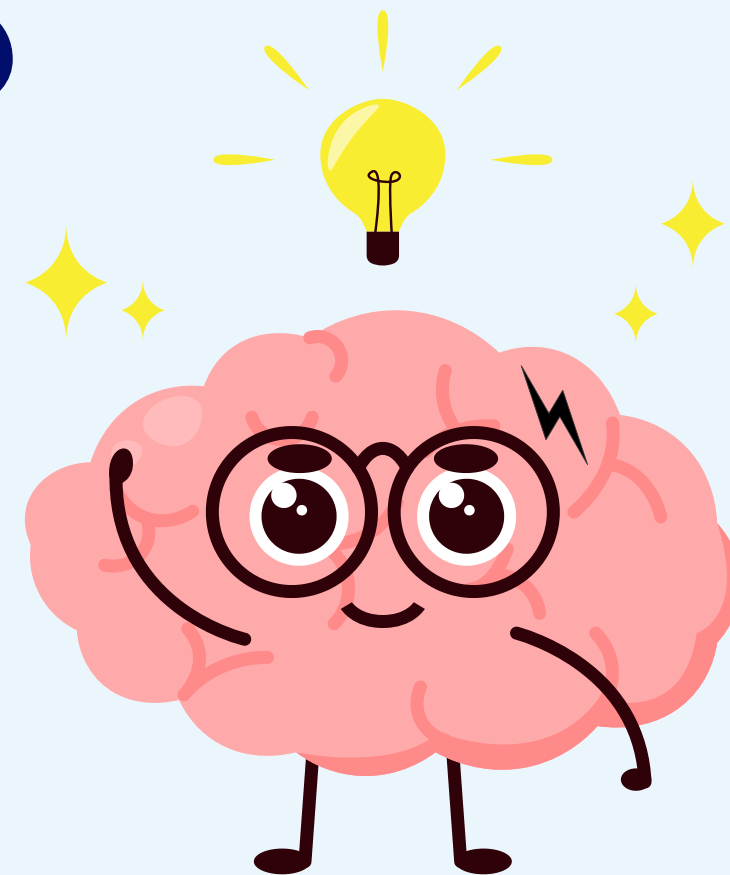


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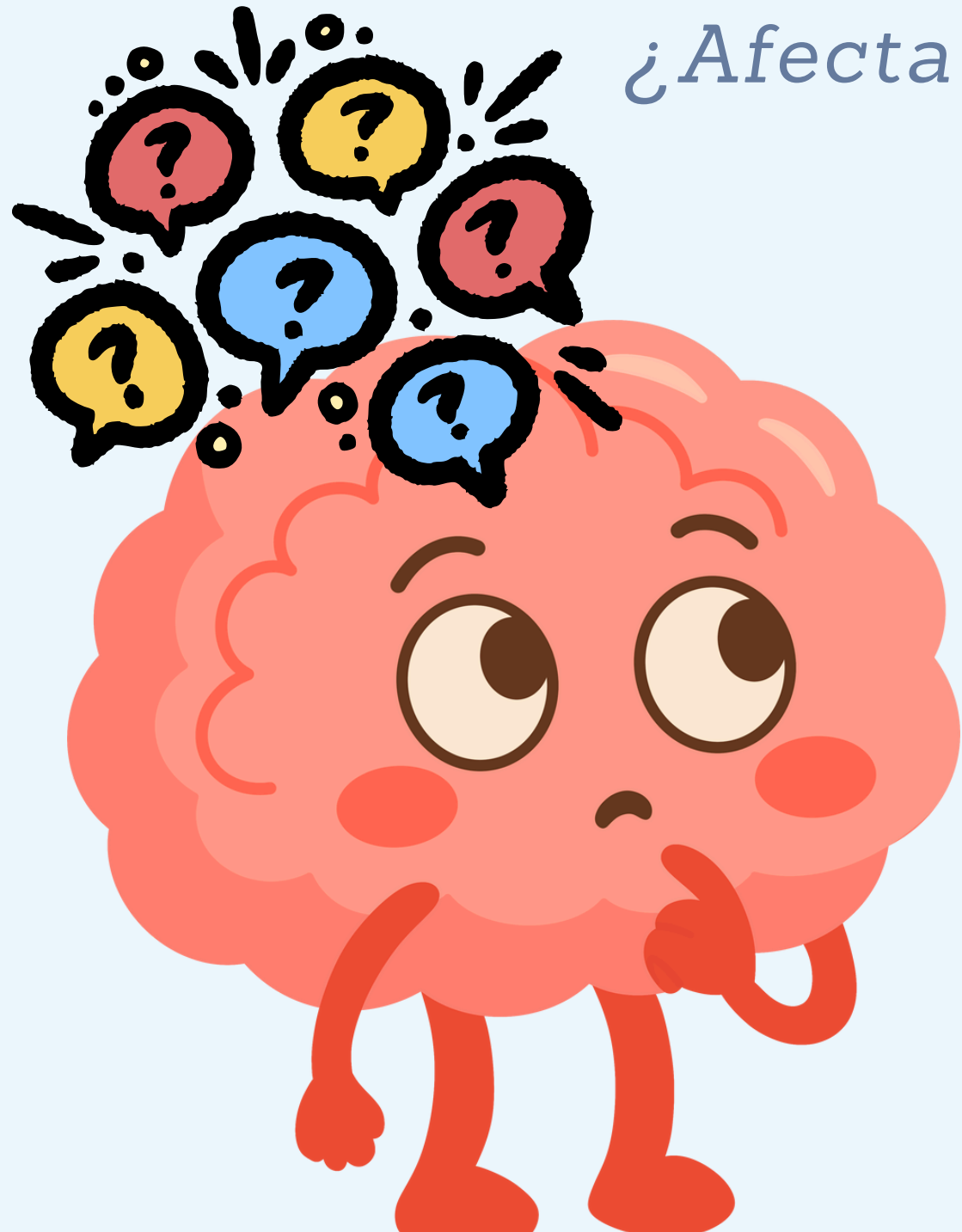
Elección de tema



**¿Sientes ansiedad
cuando usas tus redes
sociales?**



Objetivo



¿Afecta realmente el uso de redes sociales al estrés?

¿Y a la productividad?

Base de datos

	age	gender	job_type	daily_social_media_time	social_platform_preference	number_of_notifications	work_hours_per_day
0	56	Male	Unemployed	4.2	Facebook	61	6.8
1	46	Male	Health	3.2	Twitter	59	9.2

Filas

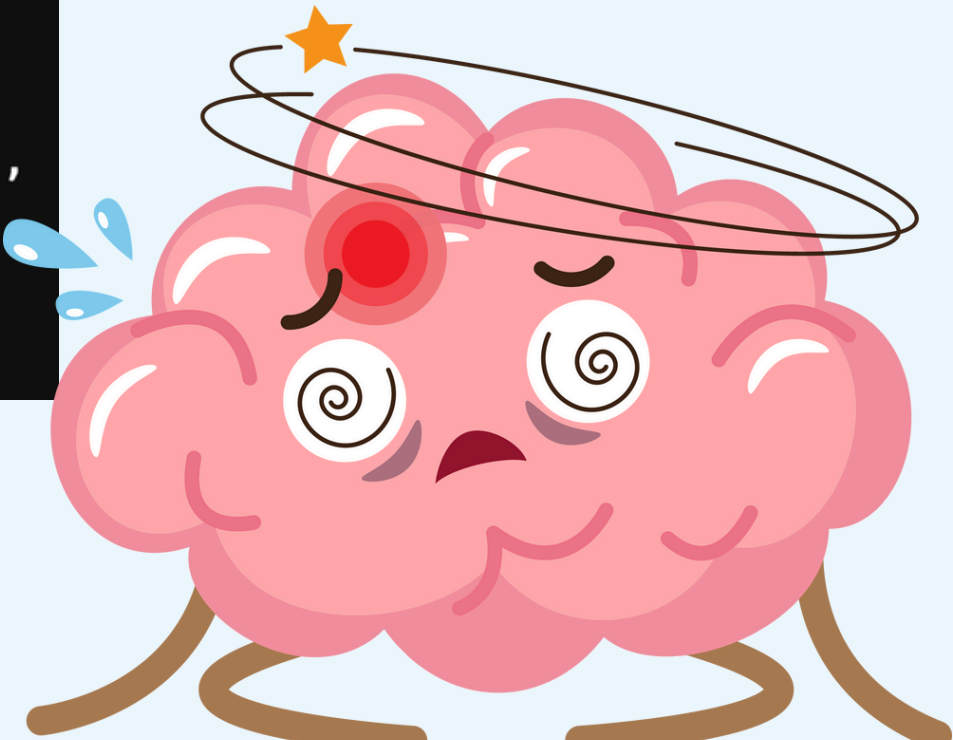
(30000, 21)

Columnas

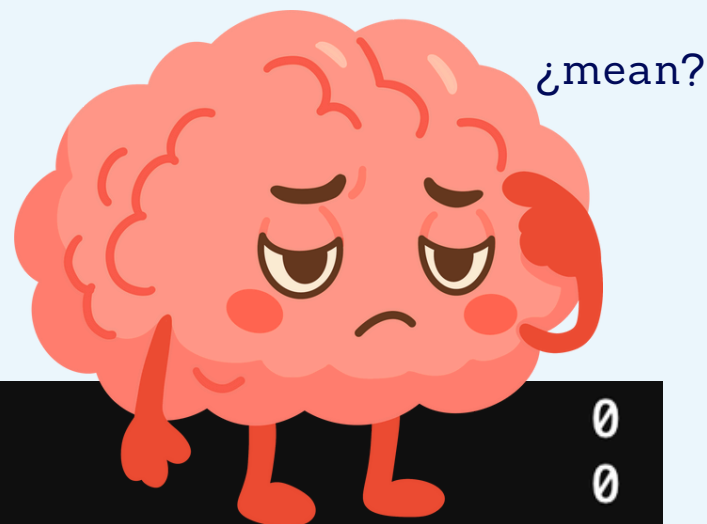
```
Index(['age', 'gender', 'job_type', 'daily_social_media_time',  
      'social_platform_preference', 'number_of_notifications',  
      'work_hours_per_day', 'perceived_productivity_score',  
      'actual_productivity_score', 'stress_level', 'sleep_hours',  
      'screen_time_before_sleep', 'breaks_during_work', 'uses_focus_apps',  
      'has_digital_wellbeing_enabled', 'coffee_consumption_per_day',  
      'days_feeling_burnout_per_month', 'weekly_offline_hours',  
      'job_satisfaction_score'],  
      dtype='object')
```



Columnas

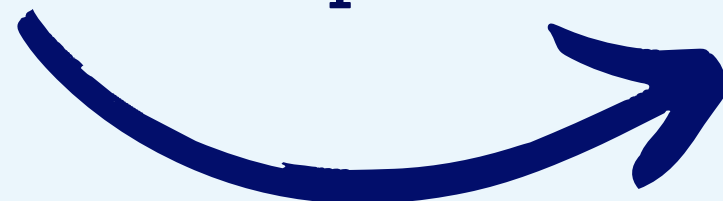


Limpieza de datos



age	0
gender	0
job_type	0
daily_social_media_time	2765
social_platform_preference	0
number_of_notifications	0
work_hours_per_day	0
perceived_productivity_score	1614
actual_productivity_score	2365
stress_level	1904
sleep_hours	2598
screen_time_before_sleep	2211
breaks_during_work	0
uses_focus_apps	0
has_digital_wellbeing_enabled	0
coffee_consumption_per_day	0
days_feeling_burnout_per_month	0
weekly_offline_hours	0
job_satisfaction_score	2730
dtype:	int64

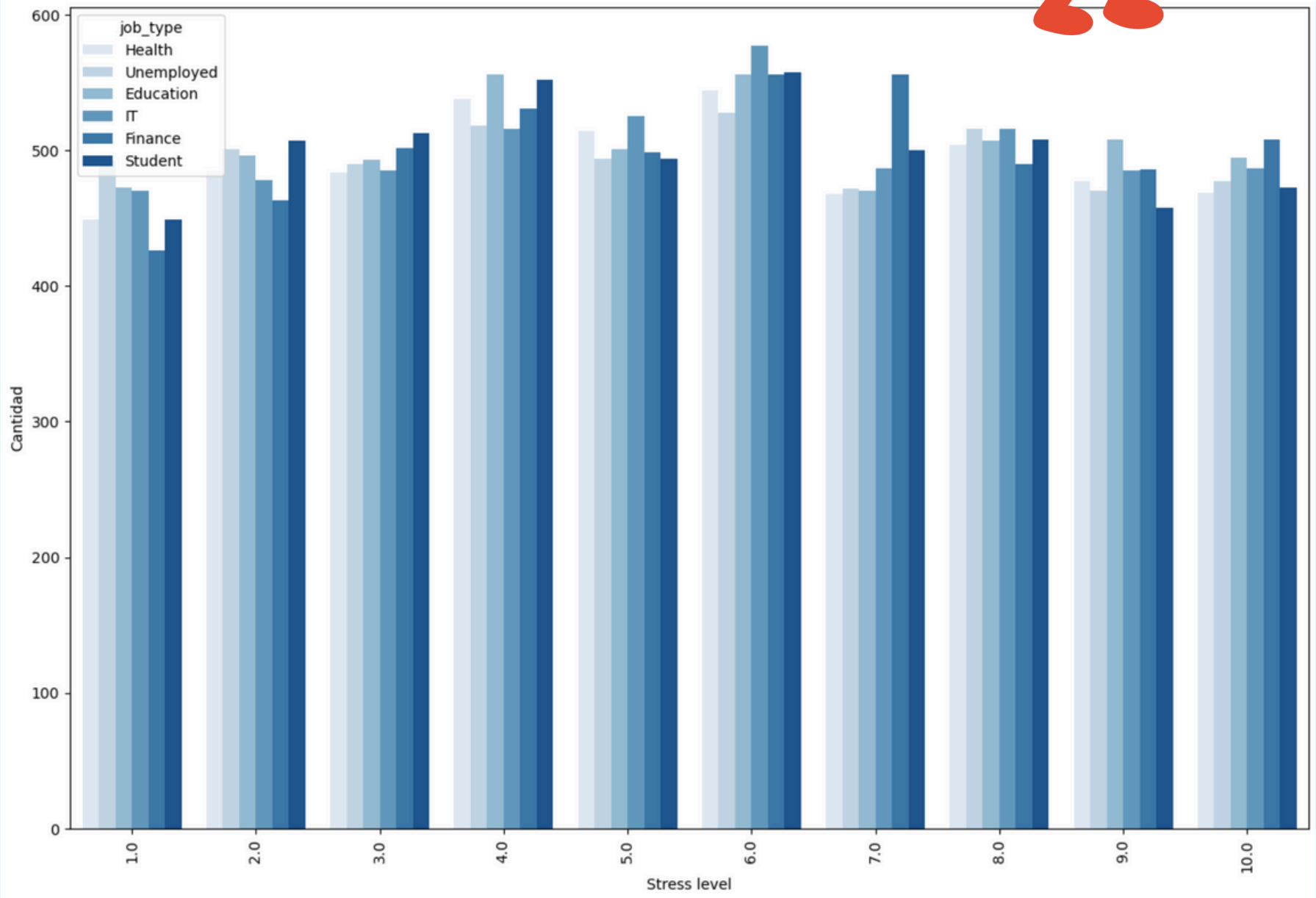
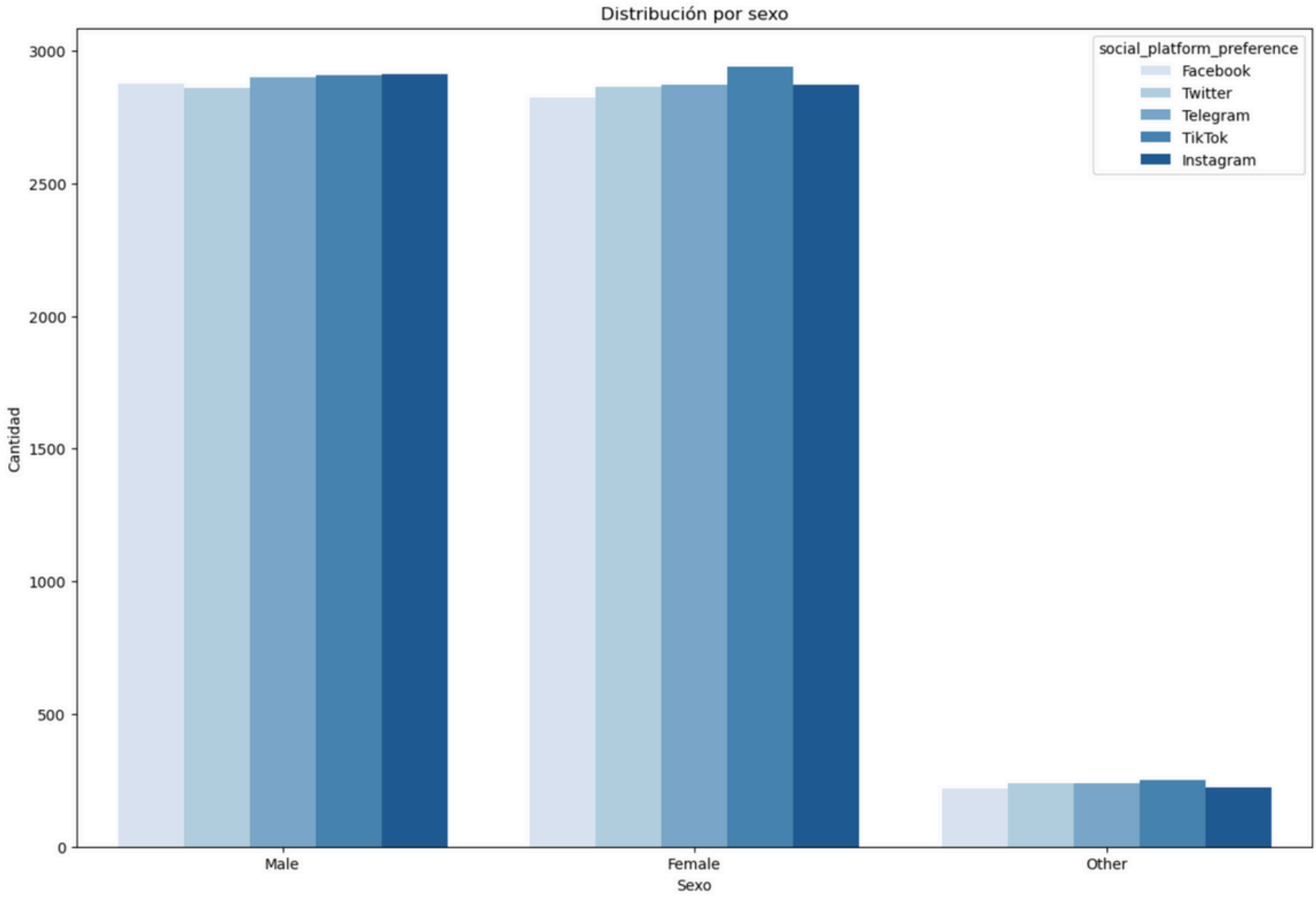
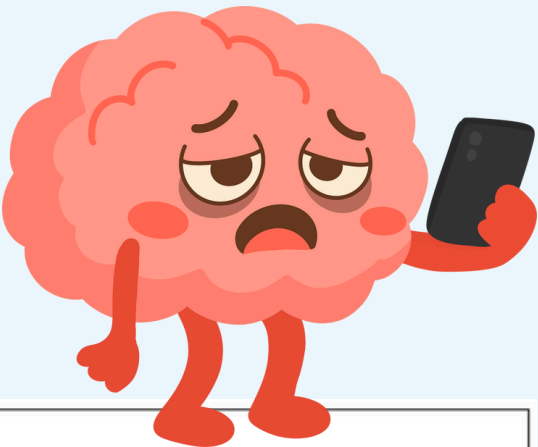
interpolate



age	0
gender	0
job_type	0
daily_social_media_time	0
social_platform_preference	0
number_of_notifications	0
work_hours_per_day	0
perceived_productivity_score	0
actual_productivity_score	0
stress_level	0
sleep_hours	0
screen_time_before_sleep	0
breaks_during_work	0
uses_focus_apps	0
has_digital_wellbeing_enabled	0
coffee_consumption_per_day	0
days_feeling_burnout_per_month	0
weekly_offline_hours	0
job_satisfaction_score	0
dtype:	int64

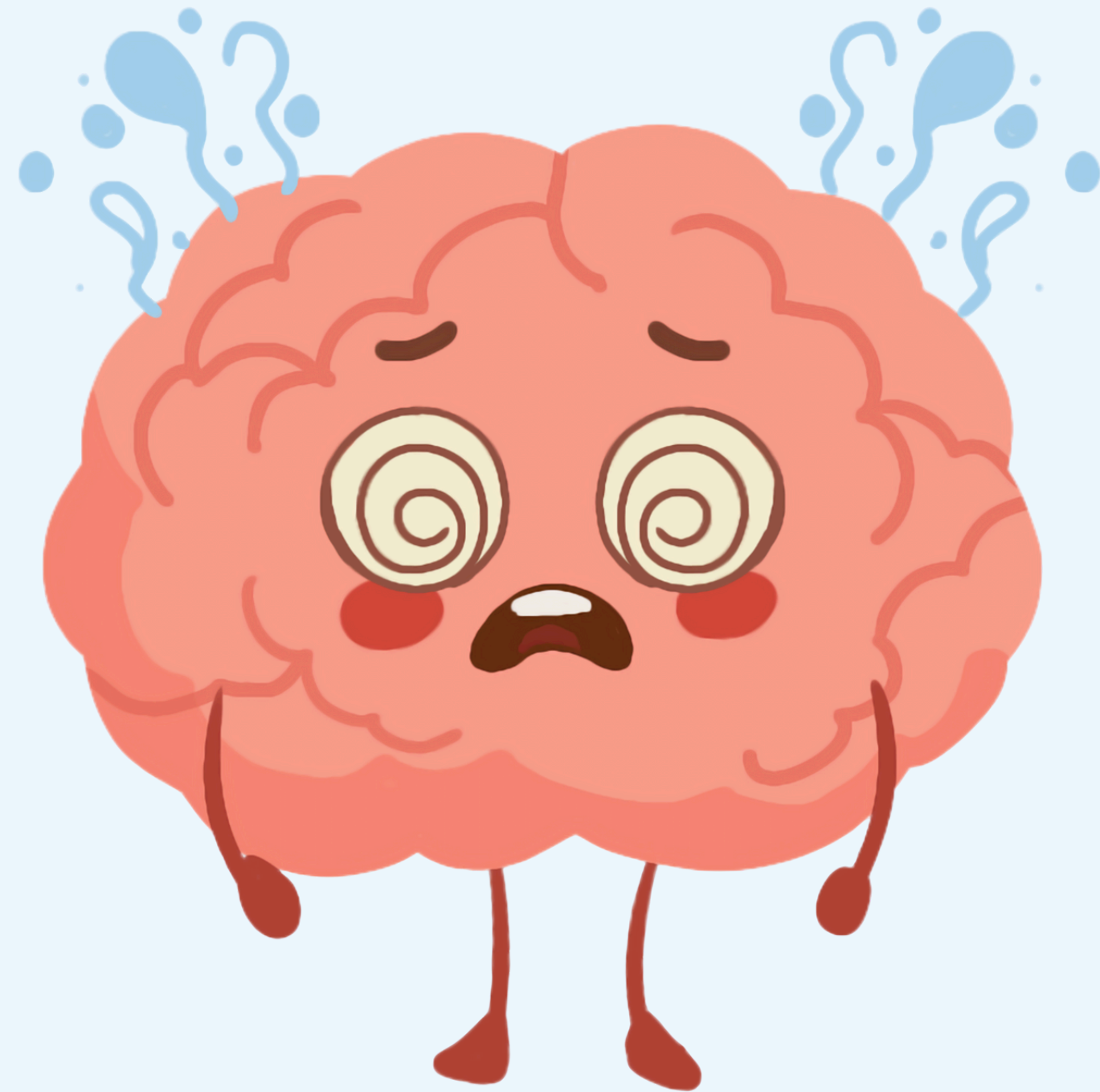


Análisis exploratorio



Respuestas

¿Hay relación con el estrés?

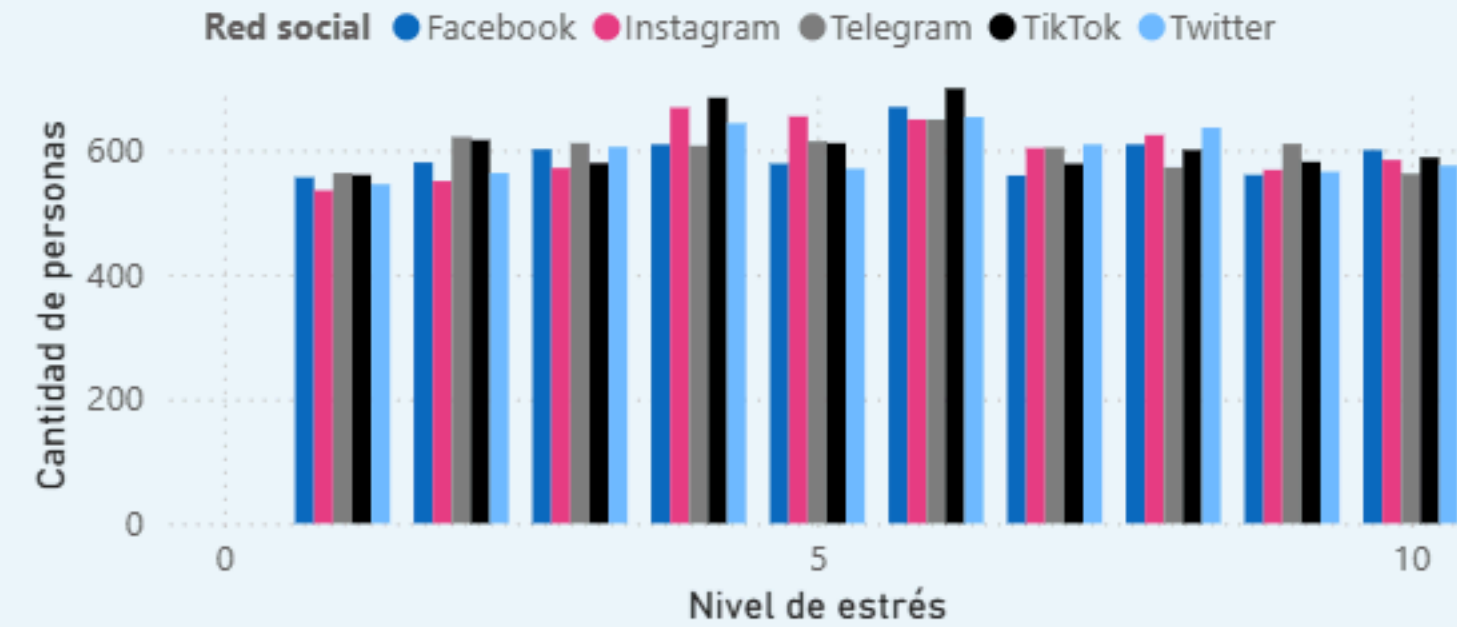


Social Media <--> Estrés

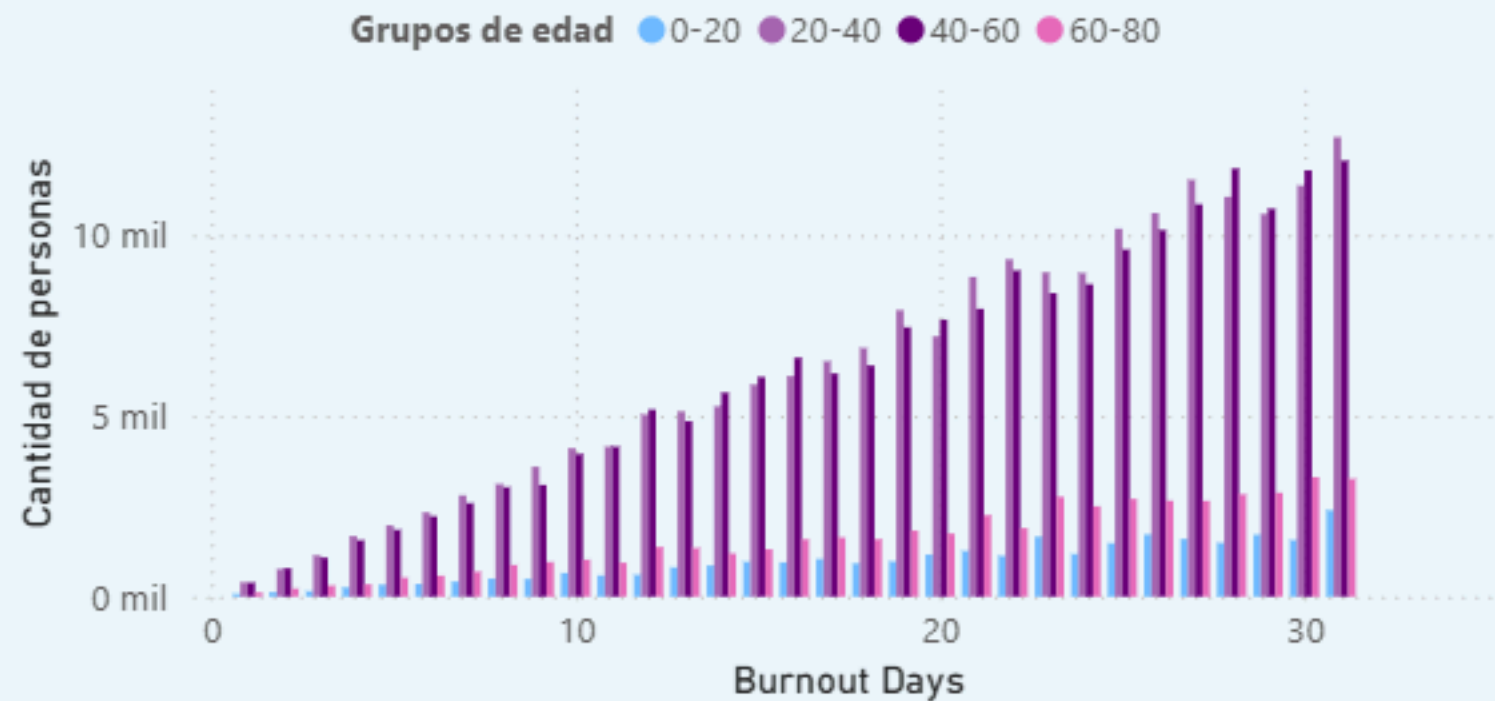
Cantidad de personas por nivel de estrés



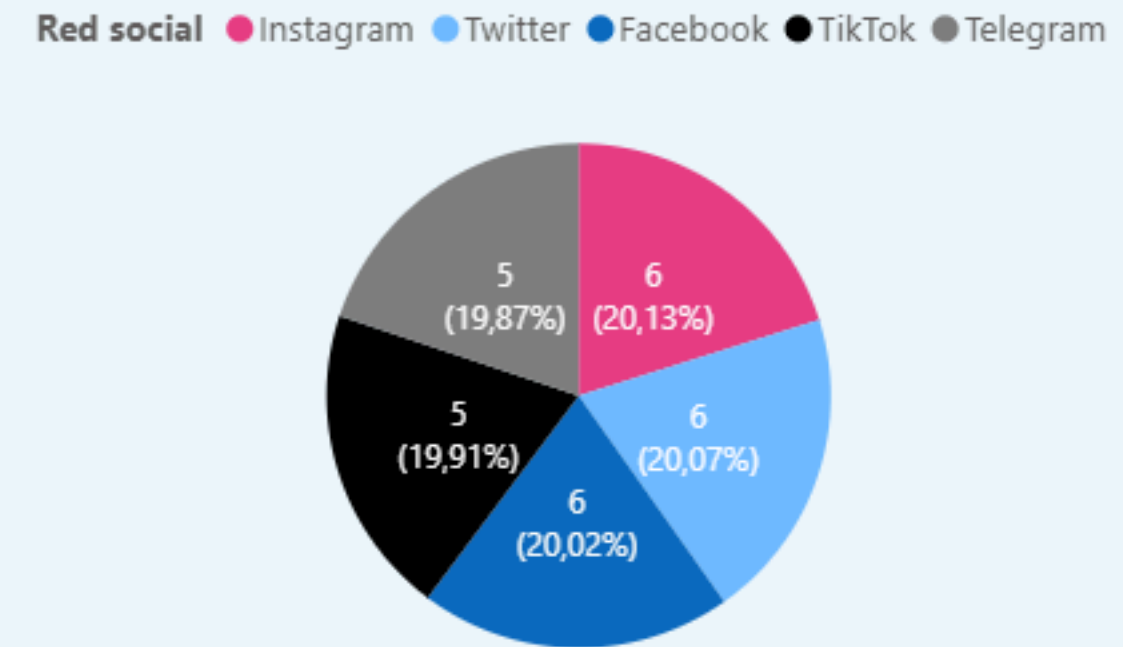
Nivel de estrés por Red social



Burnout Days por Grupos de edad

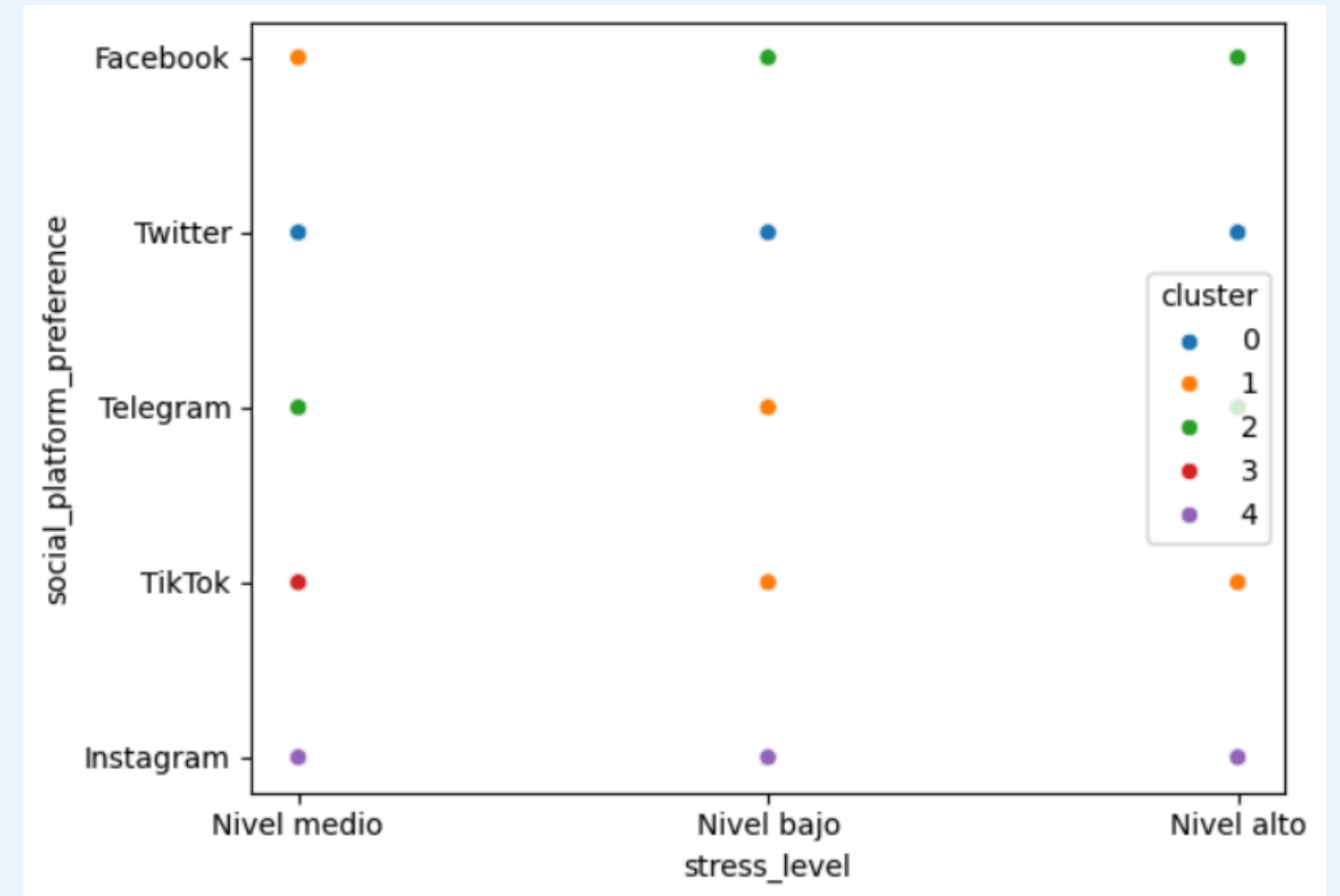
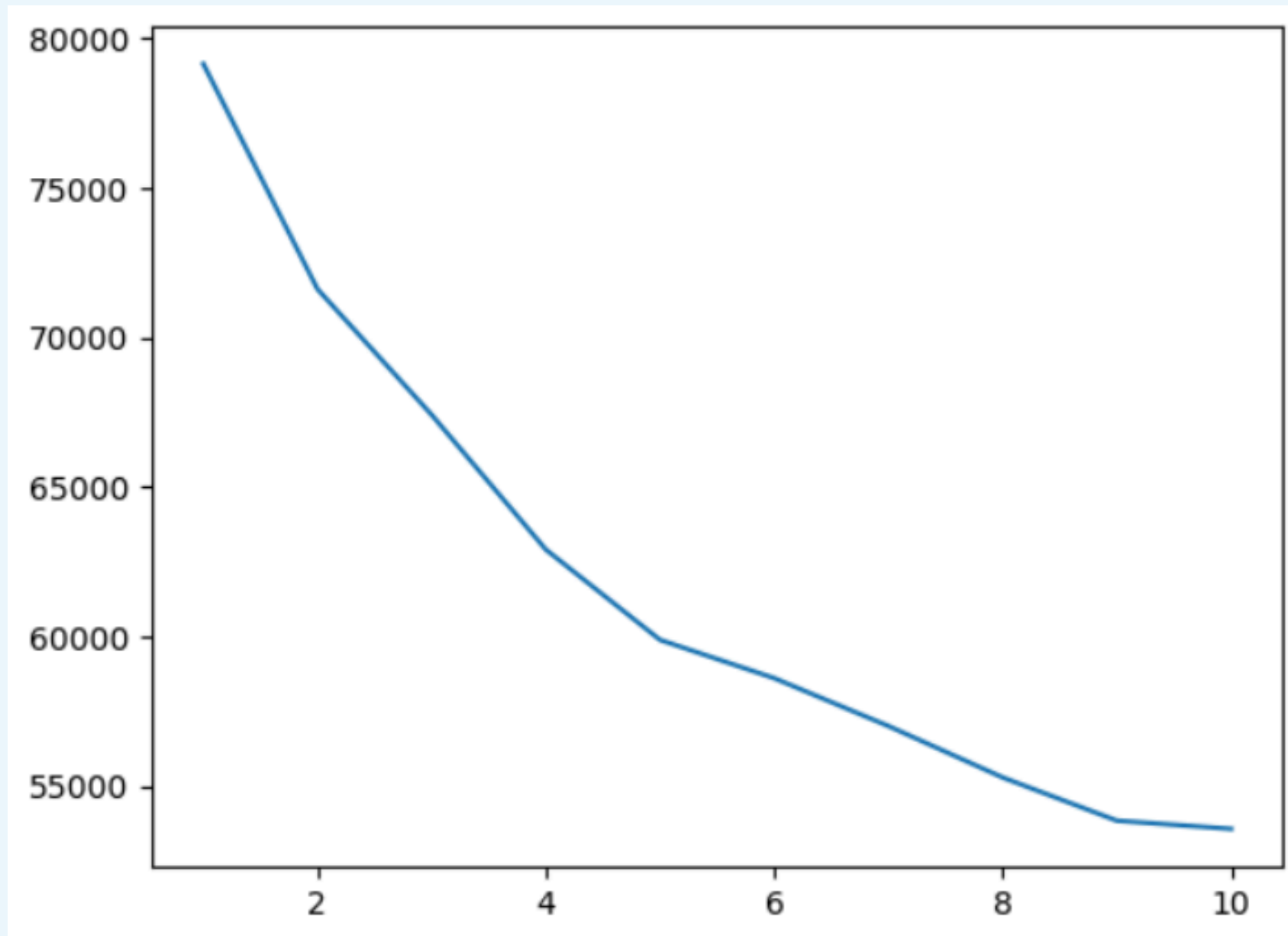


Promedio de nivel de estrés por Red social



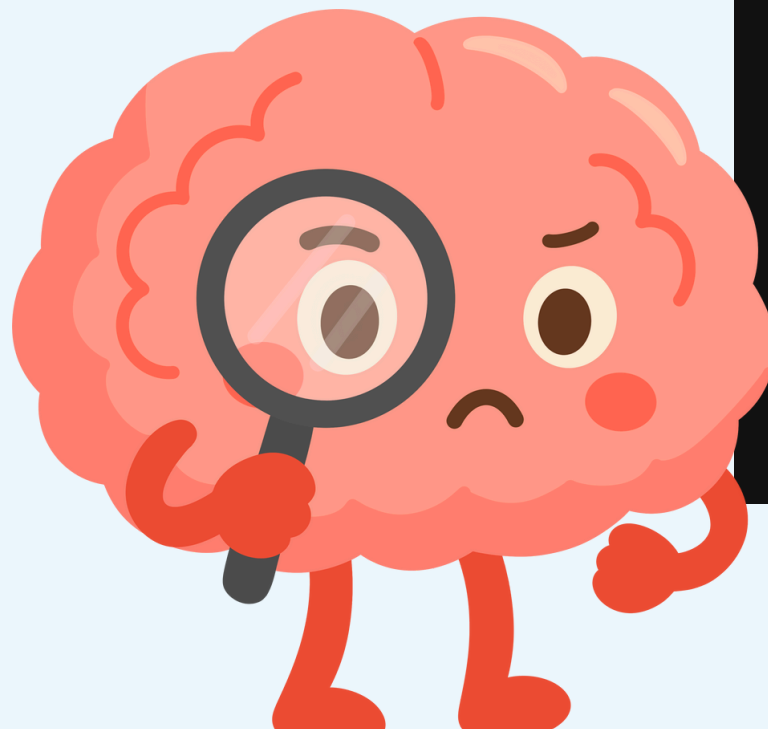
Análisis predictivo

K-MEANS



Análisis predictivo

RANDOM FOREST



```
1 -> 0.5015, 0.8133
2 -> 0.4969, 0.7404
3 -> 0.4948, 0.6635
4 -> 0.4856, 0.6764
5 -> 0.5017, 0.6266
6 -> 0.4964, 0.5937
7 -> 0.4993, 0.5927
8 -> 0.4955, 0.5811
9 -> 0.5025, 0.5632
10 -> 0.4923, 0.5469
11 -> 0.5048, 0.5587
12 -> 0.4921, 0.5187
13 -> 0.5, 0.5232
14 -> 0.4932, 0.5171
15 -> 0.5009, 0.5229
16 -> 0.4945, 0.5118
17 -> 0.4977, 0.4945
18 -> 0.5021, 0.5118
19 -> 0.4988, 0.5179
```

```
TN = matriz_confusion[0][0]
TP = matriz_confusion[1][1]
FN = matriz_confusion[1][0]
FP = matriz_confusion[0][1]
```

#Accuracy

```
(TN + TP) / (TN + TP + FN + FP)
```

0.4988

#Sensibilidad

```
TP / (TP + FN)
```

0.5179041600842549

#Especificidad

```
TN / (TN + FP)
```

0.4792004321988115

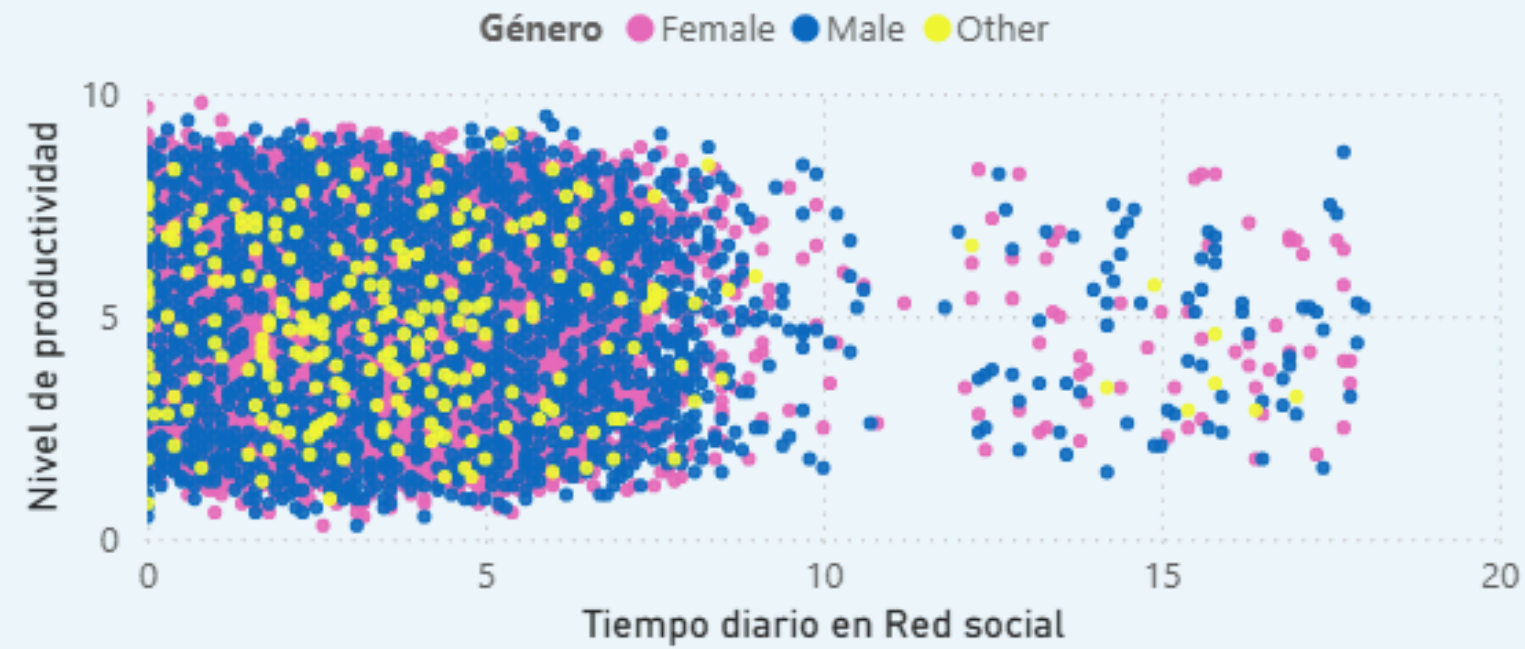
Respuestas

¿Y con la productividad?



Social Media <--> Productividad

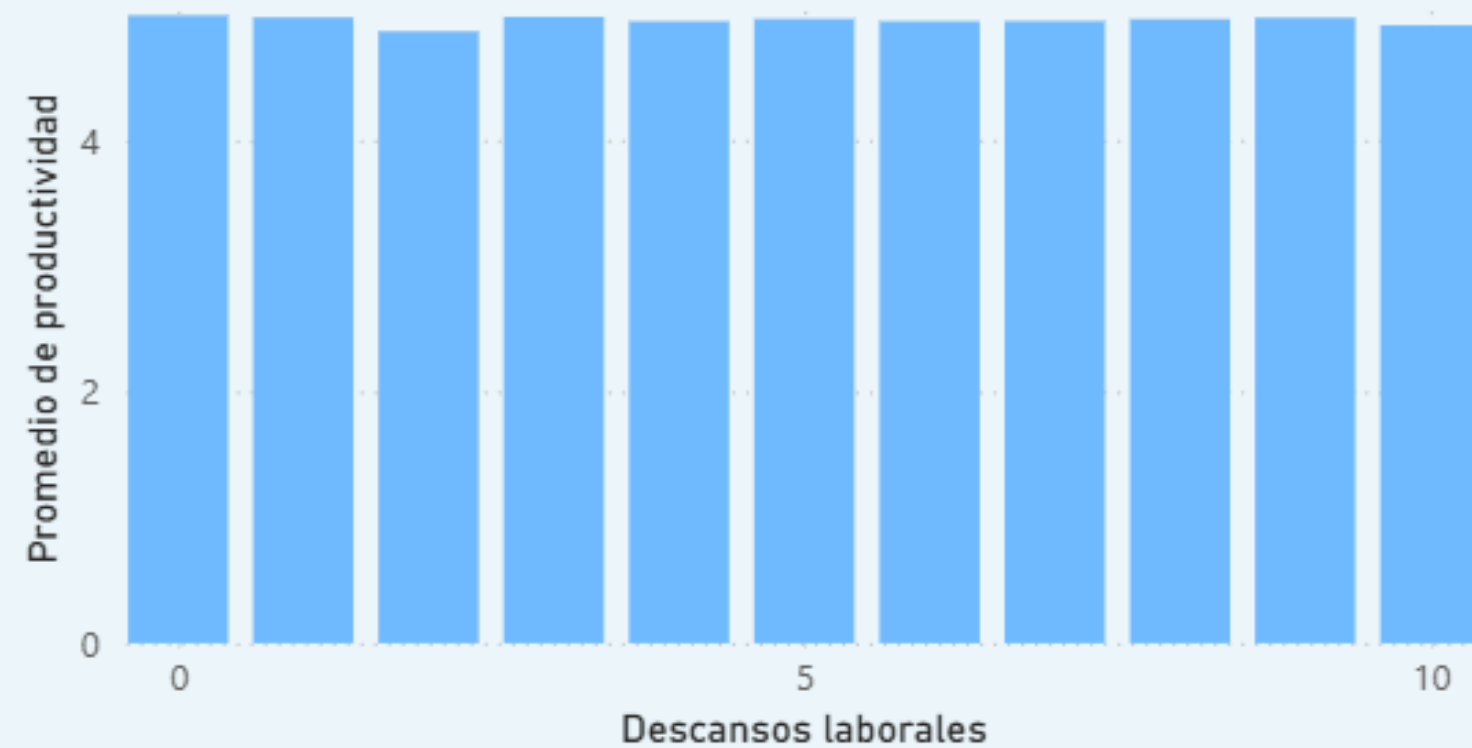
Productividad por Red social y género



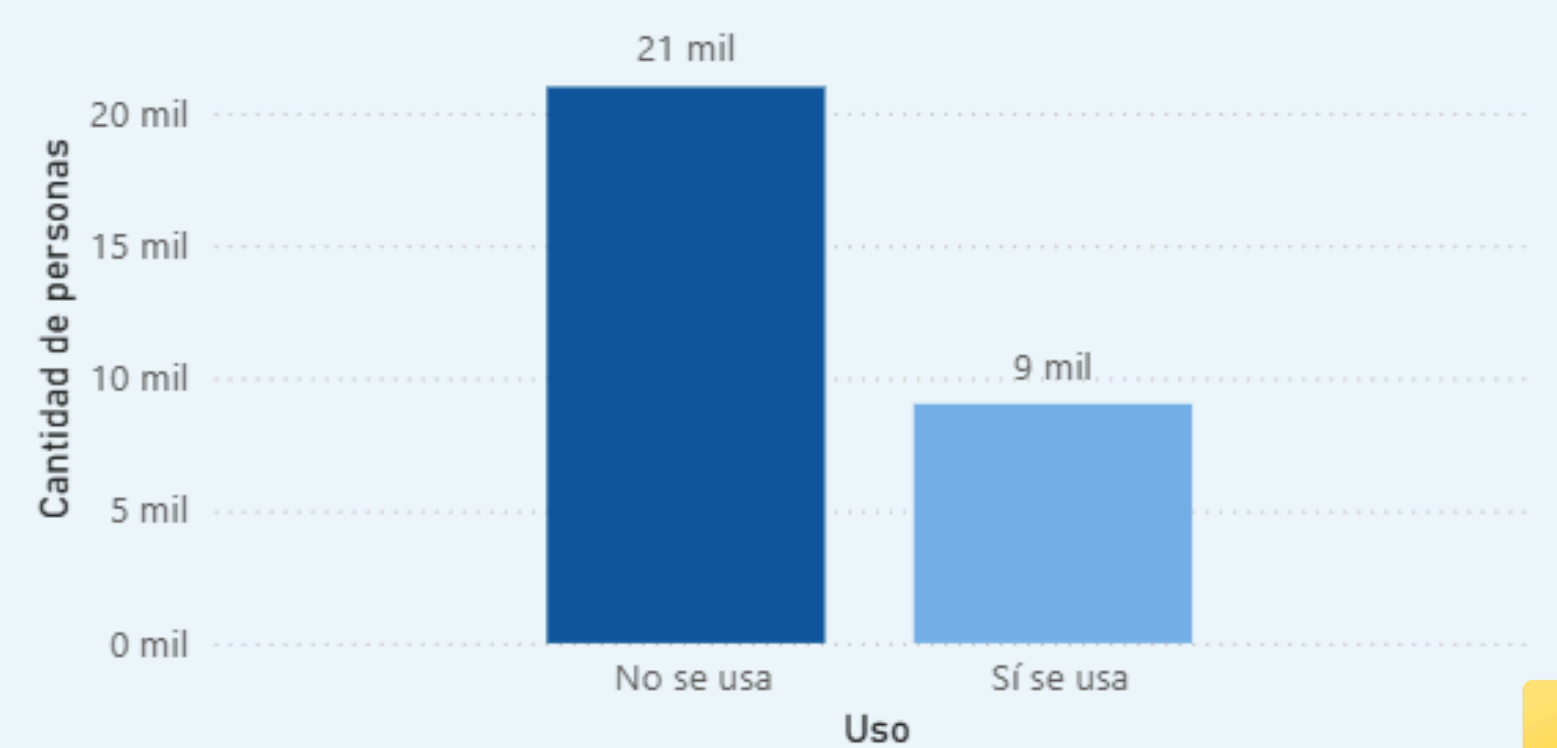
Nivel de productividad por Rango de horas



Productividad por Descansos laborales



Total de uso de Focus apps



Análisis predictivo

RANDOM FOREST

```
1 -> 0.8071, 0.784
2 -> 0.8071, 0.784
3 -> 0.8076, 0.7867
4 -> 0.8071, 0.784
5 -> 0.8068, 0.7842
6 -> 0.8069, 0.784
7 -> 0.8079, 0.7902
8 -> 0.808, 0.7897
9 -> 0.8076, 0.7861
10 -> 0.8084, 0.7872
11 -> 0.8075, 0.7867
12 -> 0.8075, 0.7867
13 -> 0.8079, 0.7913
14 -> 0.8096, 0.7932
```

#Accuracy

$(TN + TP) / (TN + TP + FN + FP)$

0.8096

#Sensibilidad

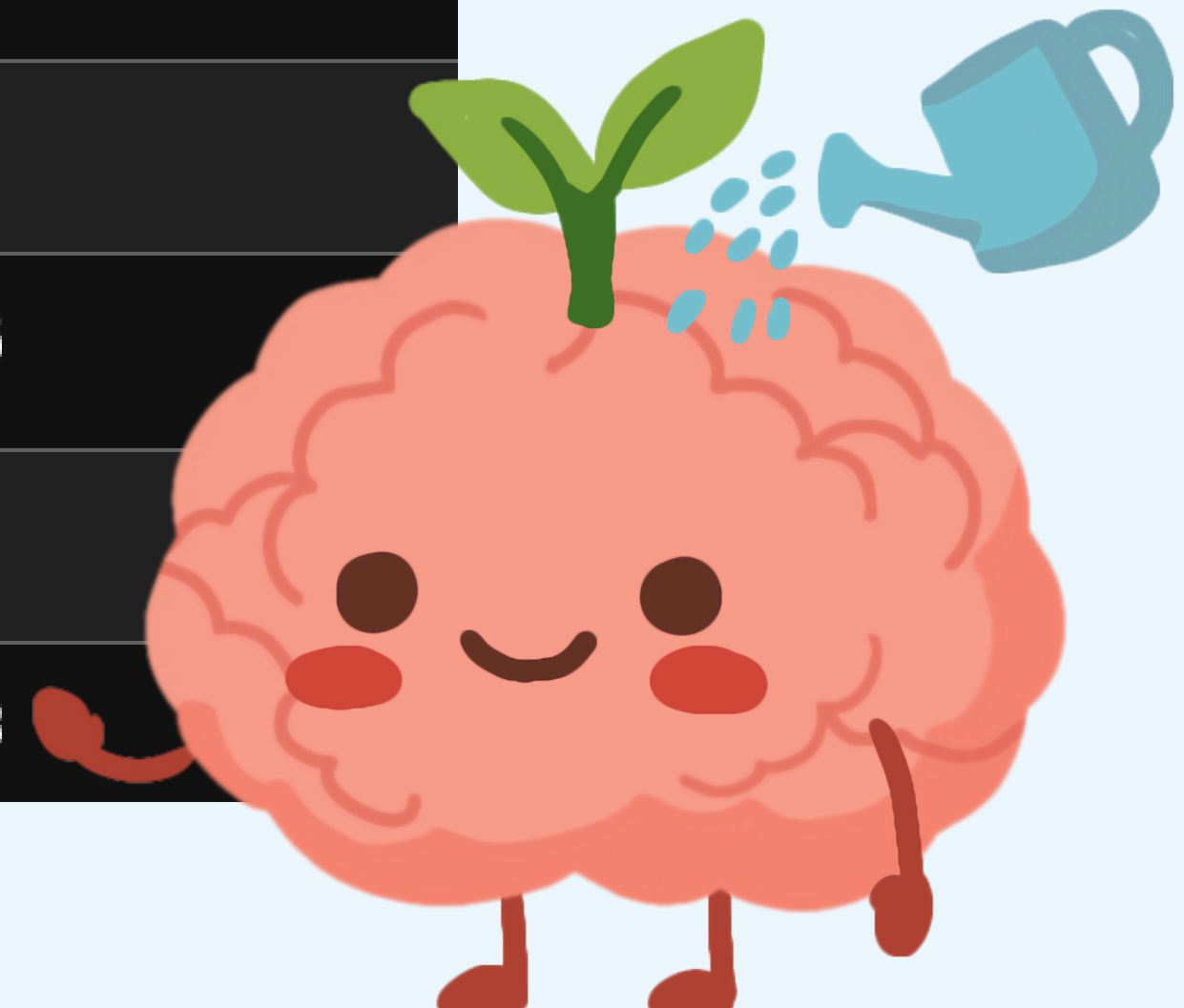
$TP / (TP + FN)$

0.7932351336606656

#Especificidad

$TN / (TN + FP)$

0.8252477829942618



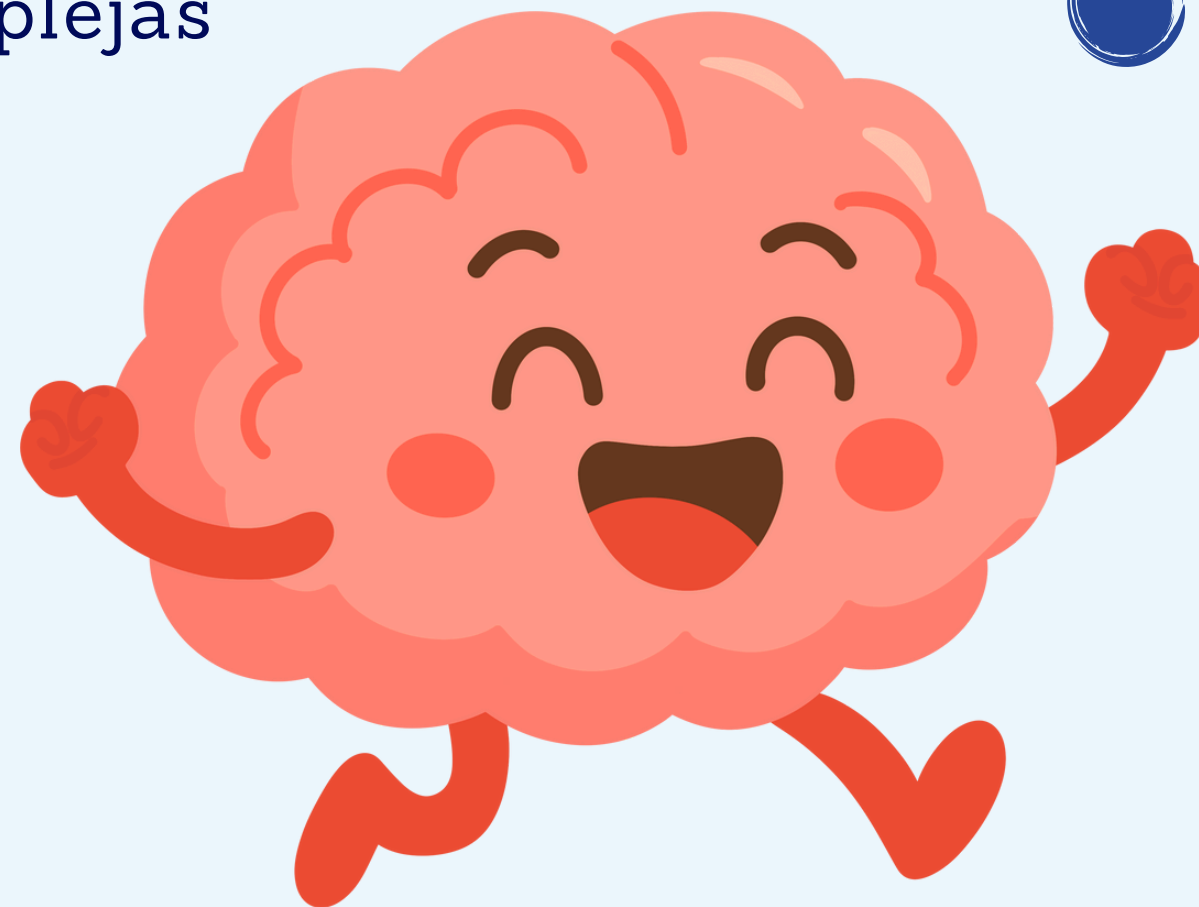
Conclusiones

Estrés

- Resultados coherentes
- Predicción aleatoria
- Variables complejas

Productividad

- Mayor relación
- Predicción más alta
- Variables más medibles



Artículos relacionados

SOCIAL MEDIA IMPACT

Social Media Usage Demonstrates Negligible Impact on Mental Well-being

 By [Press Room](#) — December 17, 2024

No Comments





JOURNAL ARTICLE

Are active and passive social media use related to mental health, wellbeing, and social support outcomes? A meta-analysis of 141 studies 🔒

[Rebecca Godard](#) ✉, [Susan Holtzman](#)

¡Muchas gracias!

